Homework 9

1. (3 pts) Determine the efficiency of a refrigerator following a Carnot cycle. It is okay to look up the answer on the internet, but put the derivation of this result in your own words.

engine efficiency

$$\eta = \frac{dW \leftarrow \text{work out}}{dQ_2 \leftarrow \text{heat in}} = \frac{dQ_2 - dQ_1}{dQ_2} = \frac{T_2 dS_2 - T_1 dS_1}{T_2 dS_2} \ *$$

1^{st} Law: $dQ_2 = dW + dQ_1$

$dW = dQ_2 - dQ_1$

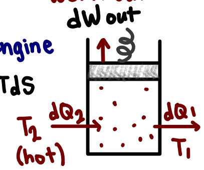
* if $dS_1 = dS_2$: $\frac{T_2 - T_1}{T_2}$ 'carnot engine' ↳ $\frac{T_H - T_C}{T_H}$

engine

work out

dW out

$dQ = TdS$



- heat flow from T_2 into system → raises pressure ($S\uparrow$)

turn off flow from T_2, piston rises → adiabatic, does work ($dW = PdV$)

- heat flow from system out T_1 → allow piston to push back (adiabatic)

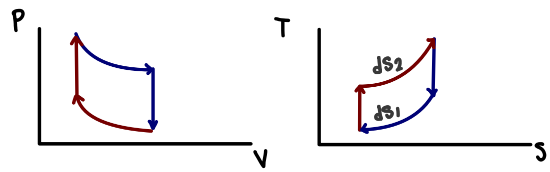
return to start

so, for a refrigerator - we know the following inversion:

engine heat → work

refrigerator work → flow from cold to hot

engine



↳ ① adiabatic compression (dW in)

→ T goes up

② connect to hot resevoir

→ gas hotter than T_2: heat flows out

③ allow it to expand by doing work on piston

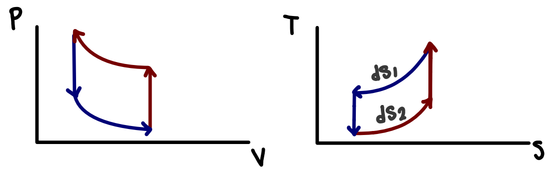
→ T drops below T_1

④ connect to T_1 → heat flow in

⇒ refrigerator efficiency:

$$\eta = \frac{dQ_1 \leftarrow \text{heat removed (via bath)}}{dW \leftarrow \text{work in}} = \frac{dQ_1}{dQ_2 - dQ_1} = \frac{T_1 dS_1}{T_2 dS_2 - T_1 dS_1} = \boxed{\frac{T_1}{T_2 - T_1} = \frac{T_C}{T_H - T_C}}$$

refrigerator



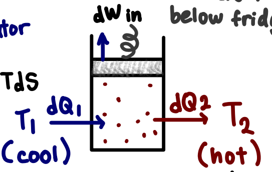
refrigerator

work in

dW in

lowers T below fridge

$dQ = TdS$



heat flows in from fridge

↳ heat can escape externally

(fridge) (compressor) (kitchen)

2. (3 pts) Calculate the heat capacity at constant volume of the phonons in a three-dimensional solid, in real units, using $E = \hbar v k$, and the Planck distribution, assuming very low temperature of $T = 1$ Kelvin, and the sound speed $v = 5 \times 10^5$cm/s. Bonus: what do you need to do to account for three different polarizations of the sound?

$$U = \frac{V}{(2\pi)^3}\int d^3k\, N_k E_k$$

where $E_k = \hbar c k$ for photons

thus, $E_k = \hbar v k$ for phonons

and planck: $N_k = \frac{1}{e^{E_k/k_BT}} = \frac{1}{e^{\hbar v k/k_BT}}$ for phonons

$\left(\frac{\pi}{L_x}\right)\left(\frac{\pi}{L_y}\right)\left(\frac{\pi}{L_z}\right) = \frac{\pi^3}{V}$

↳ volume per state in 3D k-space

where $E = \hbar v k$ for phonons

↳ $k = \frac{E}{\hbar v}$ so, $dk = \frac{dE}{\hbar v}$

$$\Rightarrow U = \frac{V}{(2\pi)^3}(4\pi)\int k^2 dk \cdot N_k \cdot E$$

$$\propto \frac{V}{2\pi^2}\int_0^\infty \left[\left(\frac{E}{\hbar v}\right)^2 \frac{dE}{\hbar v}\frac{E}{e^{E/k_BT}-1}\right] = \frac{E^2 dE}{(\hbar v)^3}\frac{E}{e^{E/k_BT}-1} = \frac{E^3 dE}{e^{E/k_BT}-1}$$

$\frac{1}{(\hbar v)^3} = \text{const.}*$

↳ $\propto \frac{V}{2\pi^2}\cdot\frac{1}{(\hbar v)^3}\int_0^\infty \left[\frac{E^3}{e^{E/k_BT}-1}dE\right]$, set $x = \frac{E}{k_BT}$: $E = xk_BT$, $dE = k_BT\,dx$

↳ ∘ $E^3 dE = (xk_BT)^3 k_BT\,dx = (k_BT)^4 x^3 dx$ *

∘ $e^{E/k_BT} = e^x$

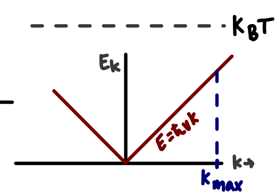
$$↳ \propto \frac{V}{2\pi^2(\hbar v)^3}(k_BT)^4\int_0^\infty \frac{x^3}{e^x - 1}dx = \frac{V}{2\pi^2(\hbar v)^3}(k_BT)^4 = \frac{(k_BT)^4}{2\pi^2(\hbar v)^3}\cdot V \ *$$

planck integral: $\frac{\pi^4}{15}$ * dimensionless ignorable!

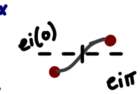
$$\Rightarrow C_V = \frac{dU}{dT} \propto T^3 \sim \frac{k_B^4 T^3}{(\hbar v^3)} k_B \cdot V \cdot \frac{*}{2\pi}$$

$U \propto (k_BT)^4$ 'Stephan-Boltzmann Law' → $C_V = \frac{dU}{dT}\Big|_V \propto T^3$

Bonus:



* $k = \frac{2\pi}{\lambda}$, $k_{max} = \frac{2\pi}{\lambda_{max}}$



→ $U = \frac{V}{2\pi^2}\int_0^{k_{max}} k^2 dk\, N_k E_k$

⇒ taking into account the 3 phonon polarities adds additional possible densities of states @ k_{max}, giving us $k_{max,1,2,3}$ and adds 3N modes since V is assumed constant

Such that:

$$C_V = 3\frac{dU}{dT}\Big|_V \propto 3T^3 \sim 3\frac{k_B^4 T^3}{(\hbar v^3)}$$

↳ if the polarities had different speeds:

$$\frac{C_V}{V} \approx \frac{k_B^4 T^3}{(\hbar v)^3}\left(\frac{1}{v_1^3} + \frac{1}{v_2^3} + \frac{1}{v_3^3}\right)$$

3. Relativity review: (4 pts)

a) In the laboratory rest frame, we see two objects of equal mass m, one traveling at 0.9c to the left, and the other traveling at 0.9c to the right. They ram into each other and become one object. Momentum conservation says that in our rest frame the final object has zero speed. What does energy conservation say that the mass of the final object will be?

energy conservation:

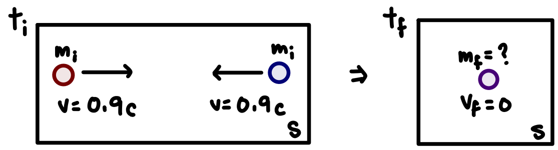
$E_F = E_i$

where $E = \gamma mc^2$

and $\gamma = \dfrac{1}{\sqrt{1-v^2/c^2}} = \dfrac{1}{\sqrt{1-(0.9c)^2/c^2}}$

↳ $\dfrac{1}{\sqrt{1-0.81}} = \dfrac{1}{\sqrt{0.19}} \approx 2.294$

and rest energy = mc^2



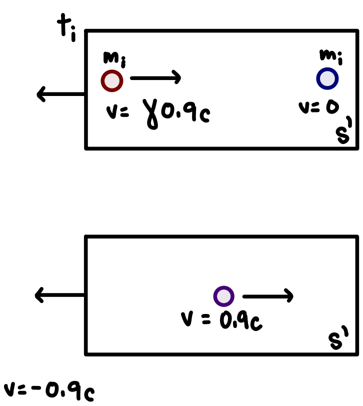
so, @ t_i: $E_{S_i} = 2\cdot\gamma m_i c^2$

and @ t_f - since $V_f = 0$: $E_{S_f} = m_f c^2$

we know: $E_{S_i} = E_{S_f} \Rightarrow m_f c^2 = 2\gamma m_i c^2$

↳ $m_f = 2\gamma m_i = 2(2.294)m_i = 4.59m$

b) Now pick the rest frame of one of the initial objects, and find the final momentum and energy of the combined object.



from the initial rest frame of the object moving to the left, the final combined object would have a relative velocity of 0.9c

where $u = \dfrac{v+u'}{1+\frac{vu'}{c^2}} = \dfrac{1.8c}{1+0.81} = \dfrac{1.8c}{1.81} = 0.9945c$

and $\gamma' = \dfrac{1}{\sqrt{1-(0.9945c)^2/c^2}} = 9.55$

so, $E_{S'_i} = \gamma' m c^2 = 9.55mc^2$

and $P_{S'_i} = \gamma' mv = (9.55)m(0.9945c) = \underline{9.497mc}$

Therefore: $E_{S'_i\,TOT} = E_{S'_f\,TOT}$ and $P_{S'_i} = P_{S'_f}$

↳ $E_{S'_i\,TOT} = E_{S'_i} + m_f c^2 = 9.55mc^2 + mc^2 = \underline{10.55mc^2}$

rest energy